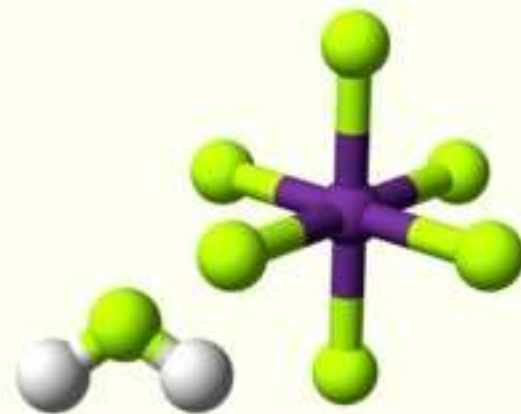


CHEMISTRY

RADIOACTIVITY

MODULE - 10



WHAT IS COMMON IN THE THREE CASES?



Smoke detectors



Emergency lights



Nuclear reactors

All these three contains radiation



WHAT IS RADIATION?



Henri Becquerel

- Radiation is the **emission or transmission of energy in the form of waves or moving stream of particles** through space or through a material medium.
- The Becquerel (Bq) is the SI unit of radioactivity.
- One Bq is defined as the activity of a radioactive material in which **one nucleus decays per second**.

Henri Becquerel discovered radioactivity by using uranium in 1896



RADIATION TYPES

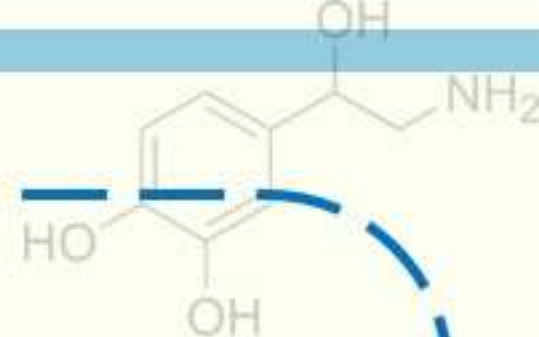
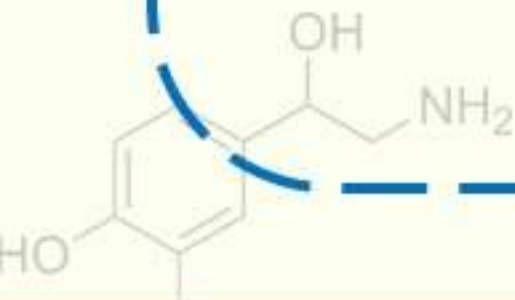
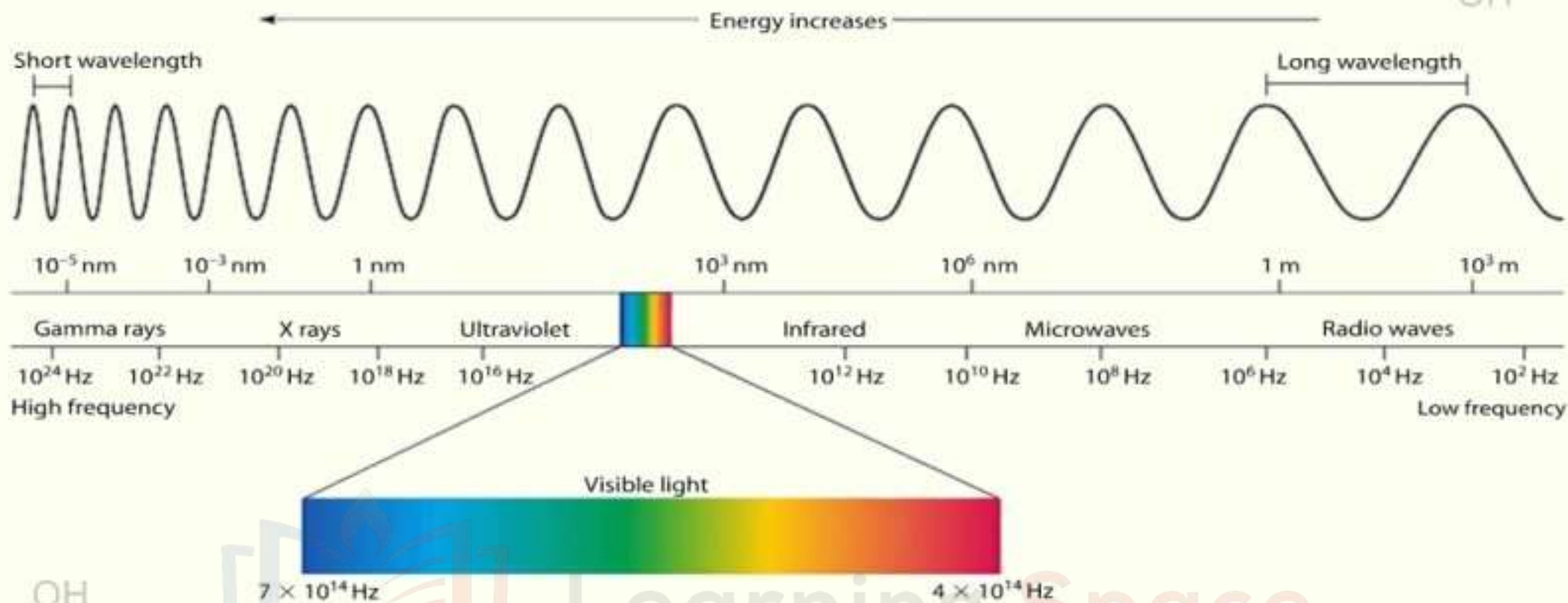
IONIZING RADIATIONS

A radiation that **has sufficient energy to remove electrons** from atoms and molecules as it passes through matter.
E.g. Gamma rays, X-rays, beta particles, alpha particles

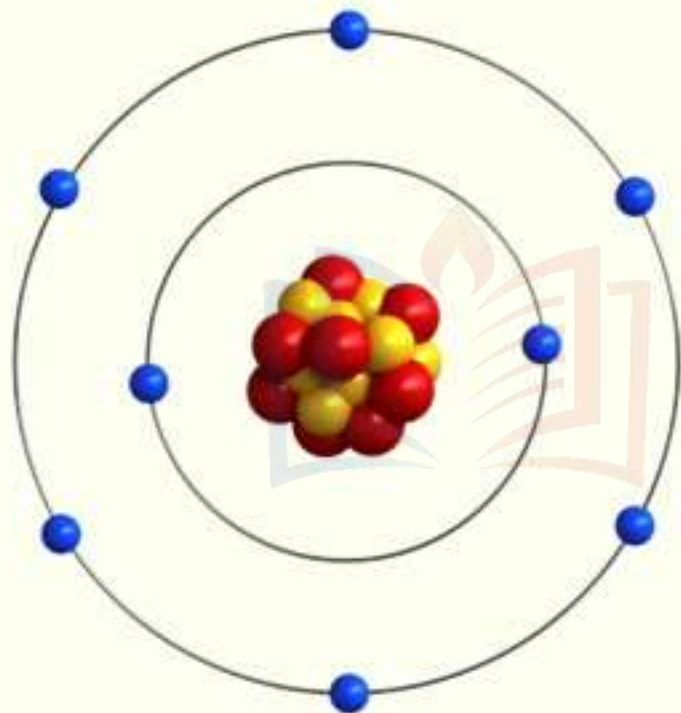
NON-IONIZING RADIATIONS

A radiation that is **not as energetic as ionizing and cannot remove electrons** from atoms or molecules.
E.G visible light, lasers, heat, microwaves and radio waves.

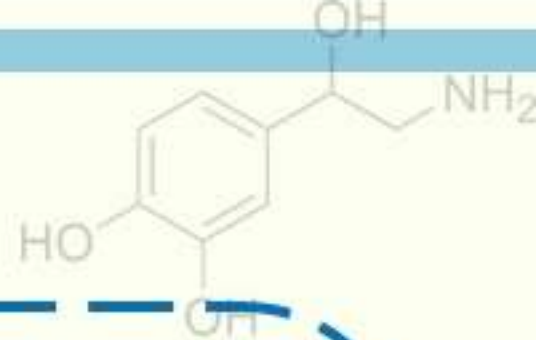
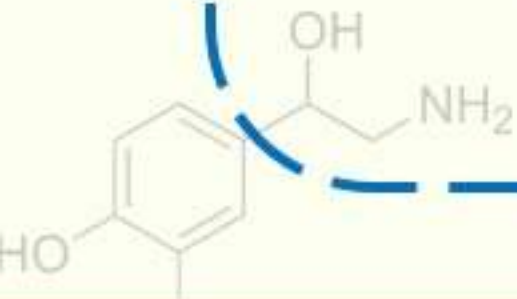




WHAT CAUSES RADIATION?



- Normally in an atom, the total number of neutrons is equal to the total number of protons.
- But, some of the elements have fewer or more neutrons
- That makes the atom unstable – radioactive
- Radioactive elements are present as radioactive isotopes.

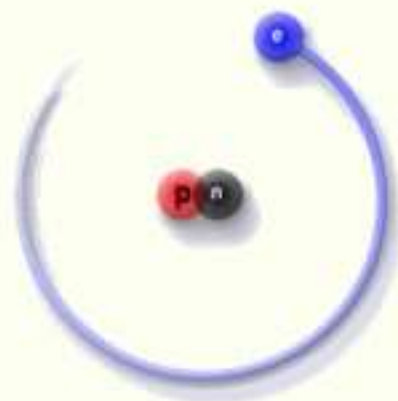


EXAMPLE OF HEAVY NUCLEUS -TRITIUM



${}^1_1\text{H}$

Protium



${}^2_1\text{H}$

Deuterium

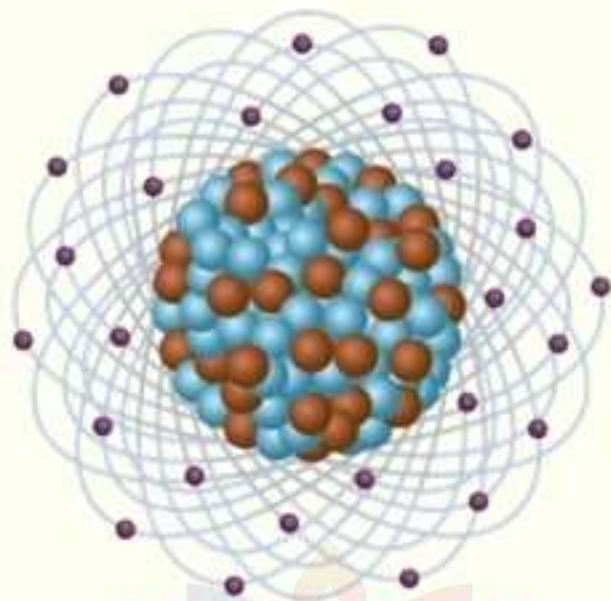


${}^3_1\text{H}$

Tritium

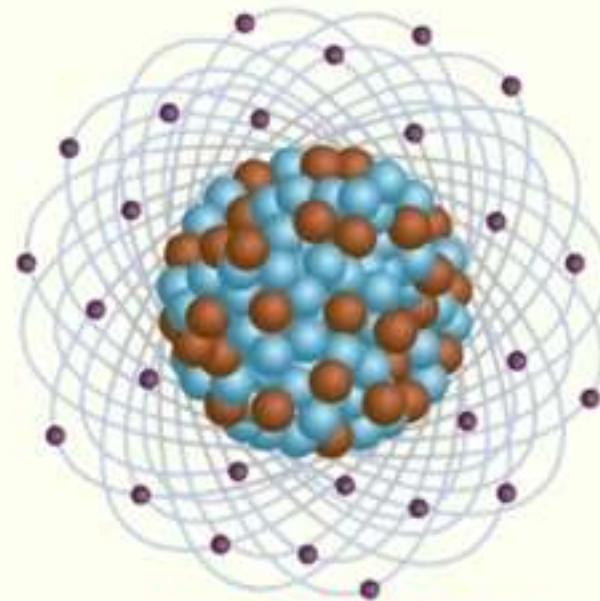


EXAMPLE OF HEAVY NUCLEUS - URANIUM



URANIUM-238

 92 protons  146 neutrons

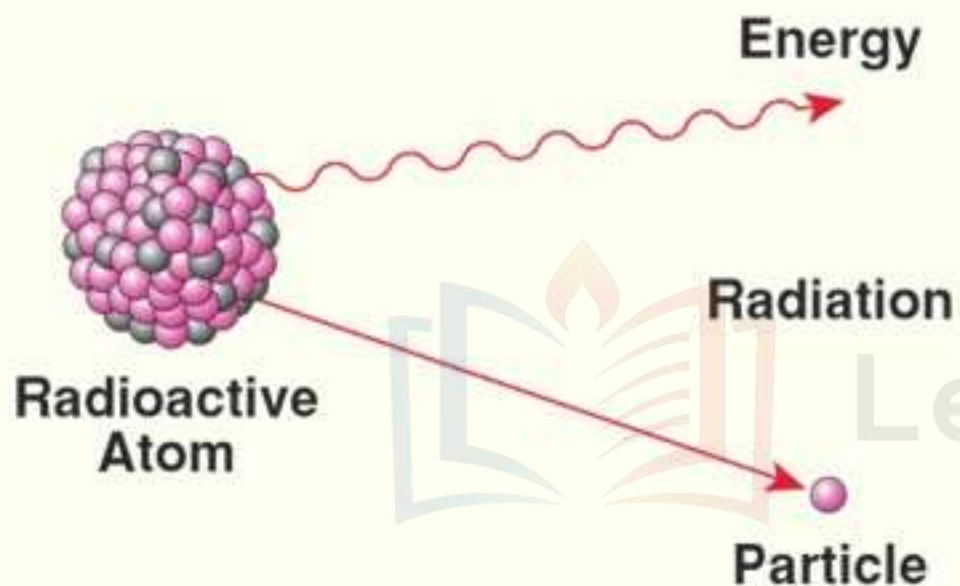


URANIUM-235

 92 protons  143 neutrons



RADIOACTIVE DECAY



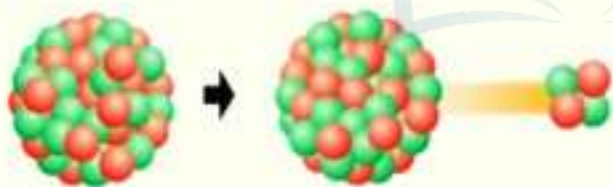
- Radioactive Decay is the spontaneous change **or** disintegration of an **unstable atomic nucleus** as it transforms itself to lose energy.
- It does this by the release of either **particulate radiation or electromagnetic radiation**, or both.

*Radioactive atoms want to become stable again
So they release energy until they get back to their balanced state*



RADIOACTIVE NUCLEAR DECAY

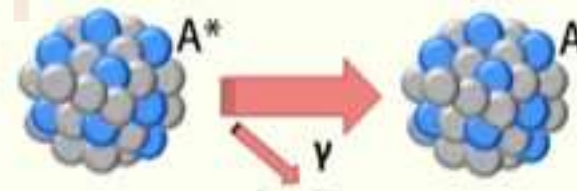
ALPHA DECAY



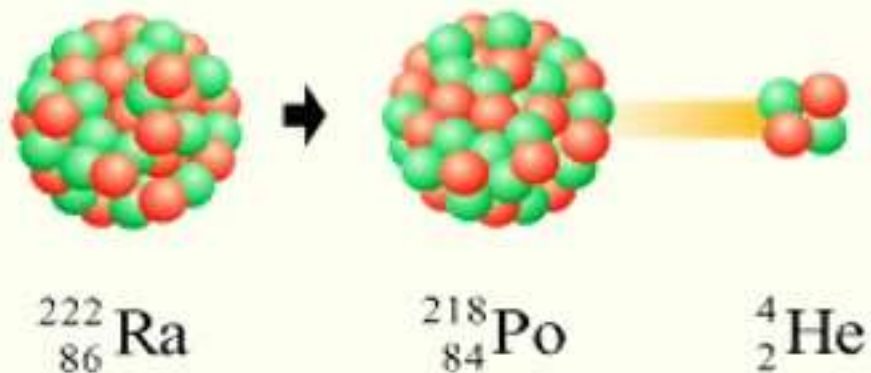
BETA DECAY



GAMMA DECAY



ALPHA DECAY



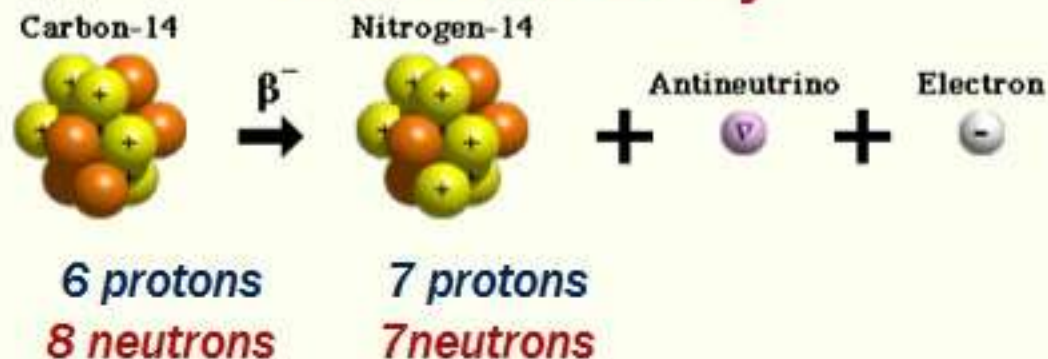
- In this process an **alpha particle** is emitted from the nucleus of an atom.
- The ejection of the alpha particle from a **nucleus** will **decrease its atomic number by two, and its mass by four.**

A radiogenic nuclide is a nuclide that is produced by a process of radioactive decay. It may itself be radioactive (a radionuclide) or stable (a stable nuclide).

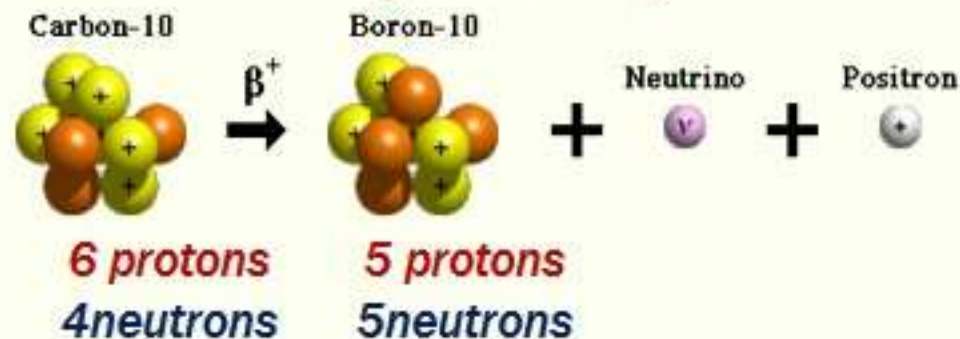


BETA DECAY

Beta-minus Decay



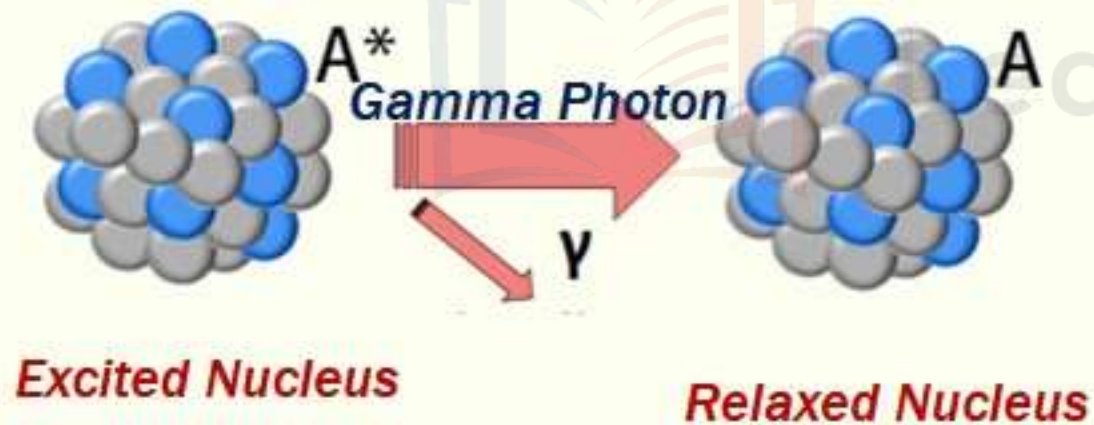
Beta-plus Decay



Beta Decay (β -decay) is a type of radioactive decay in which a proton is transformed into a neutron, or vice versa, inside an atomic nucleus.



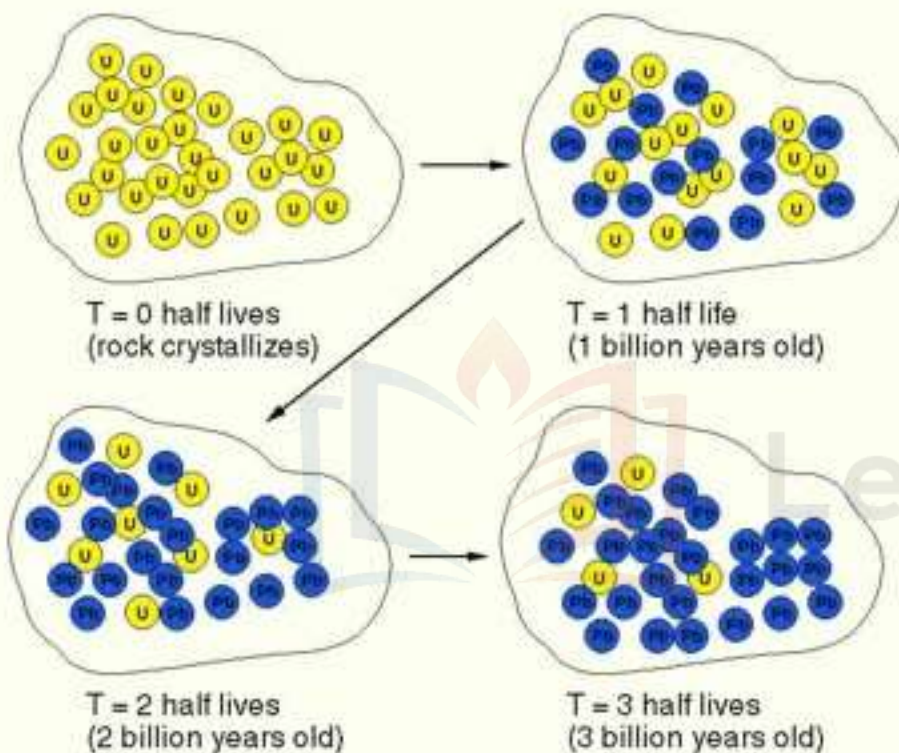
GAMMA DECAY



- Particles known as gamma rays are emitted.
- Gamma rays are very high energy photons.
- They can accompany alpha and beta decay .



RADIOACTIVE HALF LIFE



- The radioactive half-life for a given radioisotope is the time for half the radioactive nuclei in any sample to undergo radioactive decay.
- After two half-lives, there will be one fourth the original sample,
- After three half-lives one eighth the original sample, and so forth.

Half life of Francium is 22 minutes but of uranium-238 is 2.27 billion years, uranium-235 is 704 million years



URANIUM



- Uranium is extracted from an ore - Pitchblende.
- Its slow radioactive decay provides the main source of heat inside the Earth, causing convection and continental drift.
- Natural uranium as found in the Earth's crust is a mixture largely of two isotopes:
 - Uranium -238 (U-238), accounting for 99.3% and
 - Uranium -235 (U-235) about 0.7%.
- U-238 decays very slowly, its half-life being about the same as the age of the Earth (4500 million years). This means that it is barely radioactive,

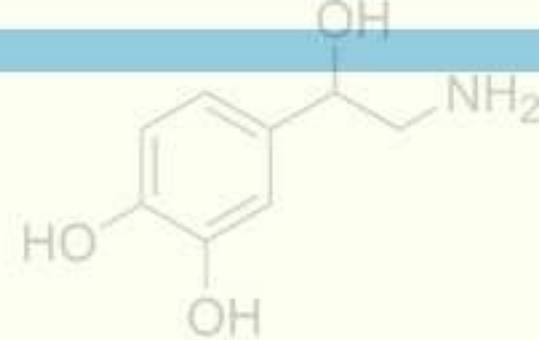
Uranium is more abundant in earth's crust as tin, tungsten and molybdenum



URANIUM

- Uranium-235 is one of the **Fissile elements**, these elements have the nucleus which fission or split, thus releasing energy. The other fissile isotope is plutonium 239.
- When the nucleus of a U-235 atom captures a moving neutron it splits in two (fissions) and releases some energy in the form of heat, also two or three additional neutrons are thrown off.
- This starts a chain reaction and it produces a very large amount of heat.
- One kilogram of uranium-235 can produce about 2×10^{13} joules of energy assuming complete fission; as much energy as 1,500 tonnes of coal.
- Whereas the U-235 nucleus is 'fissile', the U-238 is said to be '**fertile**'. It can capture one of the neutrons and become (indirectly) plutonium-239, which is fissile.





NUCLEAR REACTIONS

NUCLEAR FISSION

- Fission is the splitting of uranium atoms to release energy.
- Fission creates nuclear waste.
- Fission is more practical application at the current time

NUCLEAR FUSION

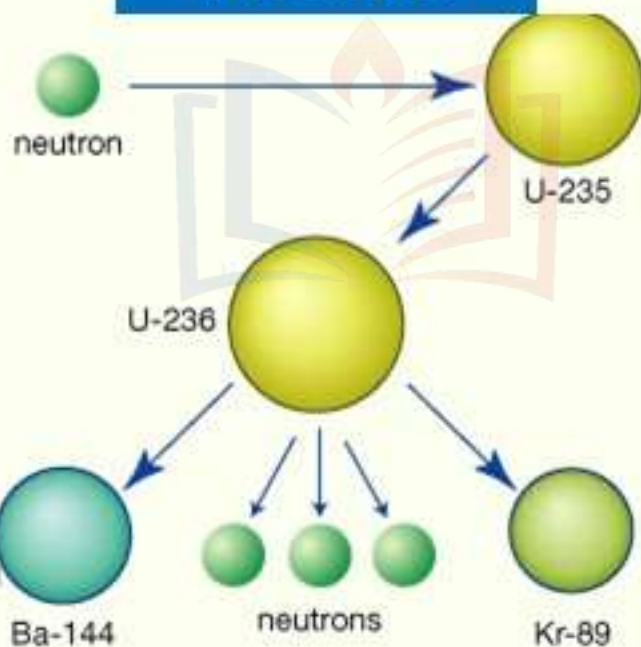
- Fusion is the combination of atoms to release energy
- Fusion has no nuclear waste
- Fusion is still being developed to be used without wasting so much energy

Nuclear fission can be controlled.

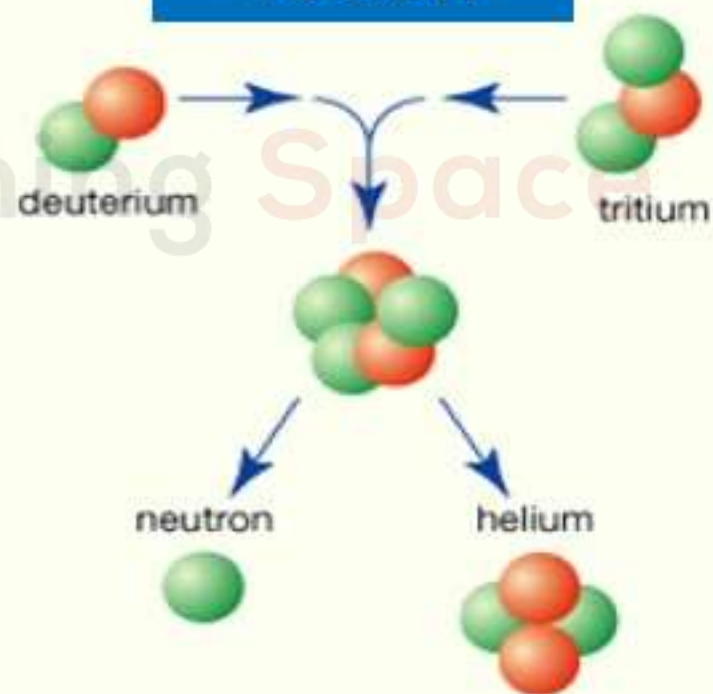


NUCLEAR REACTIONS

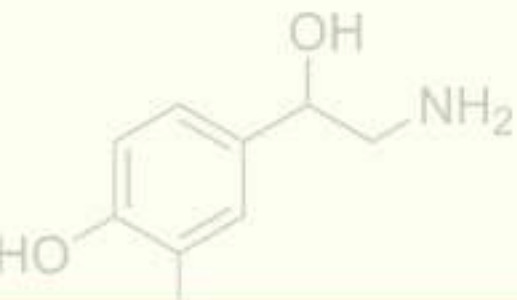
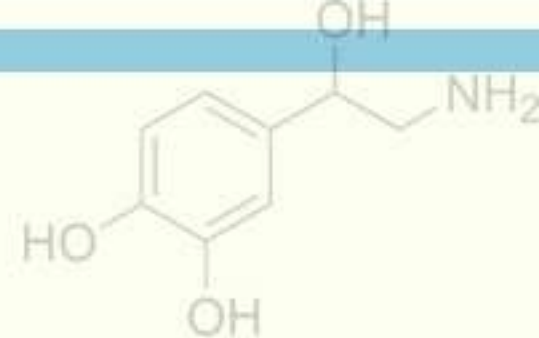
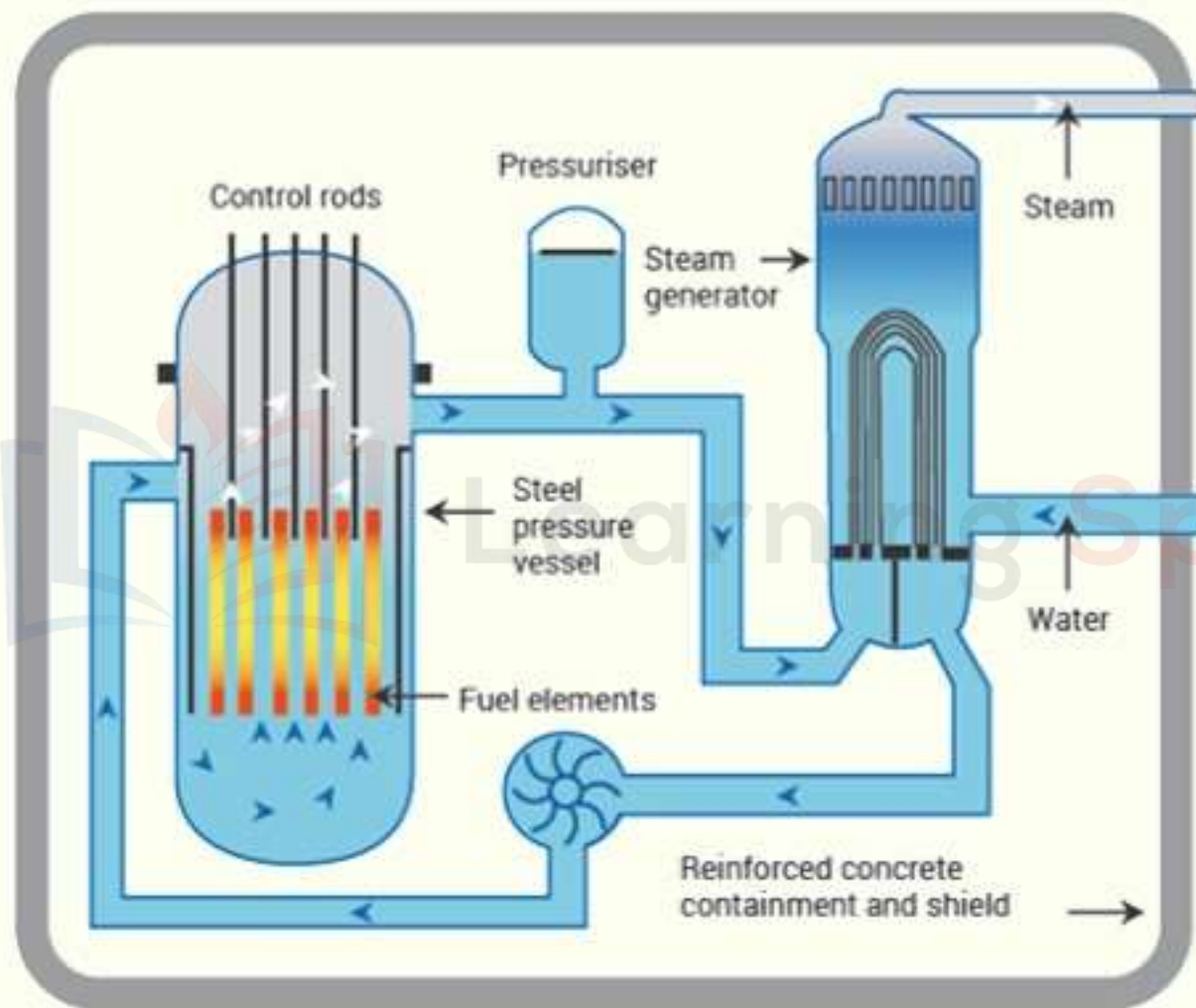
NUCLEAR FISSION



NUCLEAR FUSION



A NUCLEAR REACTOR



A NUCLEAR REACTOR

FUEL RODS

- Contain fissionable material which produces the free neutrons for chain reaction in the reactor core.
- It is usually Uranium 235

MODERATOR

- Slows down or moderate the neutrons produced when uranium or Plutonium is fissioned.
- It is usually water.

CONTROL RODS

- Absorb surplus neutrons so that the chain reaction proceeds at a steady rate and removed when the steady chain reaction reduces.
- Usually Boron or cadmium

The heat generated by nuclear fission is used to produce steam which can rotate turbines





PROCESSING THE NUCLEAR WASTE



- The spent nuclear reactor (SNF) fuel is highly radioactive initially it is too dangerous to handle.
- The radioactive material in the SNF are usually uranium and plutonium
- The un-reacted uranium and the plutonium have vastly longer half-lives, but are correspondingly less radioactive.
- Nuclear waste is stored in the pools for 20-40 years. The water cools the radioactive isotopes and shields the environment from the radiation.
- As the SNF ages the radioactivity decreases, can be placed in dry storage facilities.
- Finally they are disposed off deep underground in multi-barrier chambers.



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